coax the data to yield up an interference pattern, indicating that some of the photons actually took both paths.

Notice, too, perhaps the most dazzling result of all: the three additional beam splitters and the four idler-photon detectors can be on the other side of the laboratory or even on the other side of the universe, since nothing in our discussion depended at all on whether they receive a given idler photon before or after its signal photon partner has hit the screen. Imagine, then, that these devices are all far away, say ten light-years away, to be definite, and think about what this entails. You perform the experiment in Figure 7.5b today, recording—one after another—the impact locations of a huge number of signal photons, and you observe that they show no sign of interference. If someone asks you to explain the data, you might be tempted to say that because of the idler photons, which-path information is available and hence each signal photon definitely went along either the left or the right path, eliminating any possibility of interference. But, as above, this would be a hasty conclusion about what happened; it would be a thoroughly premature description of the past.

You see, ten years later, the four photon detectors will receive—one after another—the idler photons. If you are subsequently informed about which idlers wound up, say, in detector 2 (e.g., the first, seventh, eighth, twelfth . . . idlers to arrive), and if you then go back to data you collected years earlier and highlight the corresponding signal photon locations on the screen (e.g., the first, seventh, eighth, twelfth . . . signal photons that arrived), you will find that the highlighted data points fill out an interference pattern, thus revealing that those signal photons should be described as having traveled both paths. Alternatively, if 9 years, 364 days after you collected the signal photon data, a practical joker should sabotage the experiment by removing beam splitters a and b—ensuring that when the idler photons arrive the next day, they all go to either detector 1 or detector 4, thus preserving *all* which-path information—then, when you receive this information, you will conclude that *every* signal photon went along either the left path or the right path, and there will be no interference pattern to extract from the signal photon data. Thus, as this discussion forcefully highlights, the story you'd tell to explain the signal photon data depends significantly on measurements conducted ten years after those data were collected.